

Part 2.

In the specified code there is made use of a bitrate of 500kps, resulting in time period of 2 micro seconds per bit. In pulseview it resulted in time period of 4 micro seconds per bit. The reason for these values not matching, is due to the fact that the crystal of the MCP2515 is a 8MHz, instead of the expected 16MHz.

Part 3.

In the third part there were multiple ways to approach this assignment. The first approach is regarding to reading serial input as a string and based on the input string create an array to send out over the CAN-bus. The second approach is to read the serial input as a char array and compare each item with what has been wished for. Based on the result of this, the same array is send out over the CAN-bus. I decided to go with the first approach, since it is overall easier to read and so on easier to avoid errors.

Regarding to receiving the CAN-bus data, the incoming data is stored within a char array. Each item of this array will be checked if it matches on of the expected value combinations. Based on this the LED will turn on, off or be ignored.

Part 4.

For part 4 I had divided it into two separates parts since I only had access to 2 CAN-busses. Both of these parts were very similar so I could easily adjust the second part two their requirements. For the first part the focus was on getting the correct values for the joystick. I did not had to think too much about the data that I wanted to send over the CAN-bus since I would be using the same method as in part 3; giving each item in the char the correct value and then sending the new char over the CAN-bus to the assigned device. I the messages for per device their own message ID, to ensure only the wished for device would receive it. In the first part I also focused on adding the mask and filters to the device. The filters are there to ensure that the device would only listen to the assigned message id and therefor ignores the others. In the second part I reused the code from device A and adjusted it in such a way that it would make sense for the requirements of device B.

Since I did not had 3 CAN-busses, I researched how they should be connected to each other when using all three of them. The two 'end' nodes have the 120 ohm jumper, the middle one does not need this. Regarding to the CAN_H and CAN_L, they are all connect to their own 'sort', meaning that the CAN_H are connected to each other in one line and the same goes for the CAN_L.